

Differential Equations Shared Reference

MATH 222 H02 (Spring 2025)

1st Order Linear

Standard Form

$$y' + p(t)y = g(t)$$

Make sure to remember the standard form (including the correct signs!) to use the integrating factor properly.

2 methods to solve 1st order linear:

Integrating Factor

$$\mu = e^{\int p(t) dt}$$

where $p(t)$ is a function of t from the [standard form](#)

This is the general method. It is **guaranteed** to solve *all* 1st order linear differential equations. The next step is

$$\frac{d}{dt}(\mu y) = \mu g(t)$$

after which you simply integrate both sides and solve for y .

Separation of Variables

Integration factor fails for nonlinear cases, which is where separation of variables comes in. However, the latter is not always possible. Nonetheless, if we encounter a nonlinear equation in this class, it must be separable.

2nd Order Linear

Standard Form

$$y'' + p(t)y' + q(t)y = g(t)$$

Shortcut notation for the standard form for non-homogeneous equations:

$$L \equiv y'' + p(t)y' + q(t)y = g(t)$$

Fundamental Set of Solutions

There will be two fundamental solutions for 2nd order: y_1 and y_2

Homogeneous vs Non-homogeneous

Homogeneous when $g(t) = 0$ in the [standard form](#).

If we can solve homogeneous equations, we can solve non-homogeneous equations.

General Solution

A linear combination of the fundamental set of solutions

Homogeneous:

$$y = c_1 y_1 + c_2 y_2$$

Non-homogenous:

$$y = c_1 y_1 + c_2 y_2 + Y(t)$$

where $Y(t)$ is the particular solution.

When solving non-homogeneous equation, wait to impose IVP **after** finding Y .

Complementary Solution

For non-homogeneous equations, the part of the general solution that contains y_1 and y_2 (which are solutions to the homogeneous equation) is the complementary solution.

$$y_c = y_h = c_1 y_1 + c_2 y_2$$

Constant Coefficients

p and q are constants, NOT functions of t .

Substitute $y = e^{rt}$ into the [standard form](#), but **homogenous** form, to get something like:

$$ar^2 + br + c = 0$$

2 Distinct, Real Roots $b^2 - 4ac > 0$	$y = c_1 e^{r_1 t} + c_2 e^{r_2 t}$ End of Midterm 1 content
Repeated Roots $b^2 - 4ac = 0$	$y = c_1 e^{rt} + c_2 t e^{rt}$ (Same r) Multiply the second term by t.
Complex Roots $b^2 - 4ac < 0$	$y = e^{\lambda t} (A \cos \mu t + B \sin \mu t)$ $r = \lambda \pm \mu i$

Wronstkian

The determinant

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2$$

Existence Uniqueness Theorem

- If p and q are continuous on an interval I containing t_0 , then the IVP (initial value problem) has a unique solution on I .
- If $W(y_1, y_2)|_{t_0} \neq 0$, then $y = c_1 y_1 + c_2 y_2$ is a **unique** solution to the IVP. (It also means that y_1 and y_2 are linearly independent?)

Particular Solution

A particular solution $Y(t)$ is such that $L[Y] = g(t)$.

Thus, the general solution is $y = c_1 y_1 + c_2 y_2 + Y$.

Multiple particular solutions exist because, for example:

$$L[Y + 4y_1] = L[Y] + L[4y_1] = g(t)$$

because y_1 sums to 0. So, $Y + 4y_1$ is another particular solution.

Note: the particular solution has its own constants.

Do **NOT** add arbitrary constants when constructing the general solution.

Finding a Particular Solution

MUC - Method of Undetermined Coefficients

Method to find a particular solution by “guessing” cleverly.

Requirements:

1. Constant coefficients only!!	2. $g(t)$ must be “sufficiently simple.”
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1. Once you choose the correct form (from the right column below), differentiate it twice.
2. Substitute into the differential equation, which itself equates to $g(t)$.
3. Match e^t , $\cos t$, $\sin t$, and same degree terms with each other via factoring. Some could sum to 0.
4. Solve the system of equations to find the coefficients.

Note: for polynomials, the entire general form has to be written, e.g. $g(t) = t^3 + 2t$ means that the guess for $Y = At^3 + Bt^2 + Ct + D$.

Always solve for the complementary (homogeneous) solution first! This way, you can see if any fundamental solutions clash with $g(t)$. If **ANY** term in Y is part of the complementary solution, then **multiply the necessary terms** of Y by t .

For example, if $g(t) = t \cos 2t + 4e^{2t}$ and $y_c = c_1 \cos 2t + c_2 \sin 2t$, then

$$Y = t(A_0 t + A_1) \sin 2t + t(B_0 t + B_1) \cos 2t + Ae^{2t}.$$

Notice how the last term is NOT multiplied by t because it does NOT appear in y_c .

You may have to multiply by t **multiple times** until NO part of Y appears in y_c !

$g(t)$	$Y(t)$ (particular solution guess)
$e^{\alpha t}$	$Ae^{\alpha t}$
$\sin \beta t$ $\cos \beta t$	$A \sin \beta t + B \cos \beta t$
$a_n t^n + a_{n-1} t^{n-1} + \dots + a_0$	$A_n t^n + A_{n-1} t^{n-1} + \dots + A_0$
$(a_n t^n + \dots + a_0)e^{\alpha t}$	$(A_n t^n + \dots + A_0)e^{\alpha t}$
$e^{\alpha t}(a_n t^n + \dots + a_0) \sin \beta t$ $e^{\alpha t}(a_n t^n + \dots + a_0) \cos \beta t$	$e^{\alpha t}(A_n t^n + \dots + A_0) \sin \beta t +$ $e^{\alpha t}(B_n t^n + \dots + B_0) \cos \beta t$

MVP - Method of Variation of Parameters

The more general method. Has none of the requirements that MUC has.

The for a particular solution can be derived, resulting in:

$$Y(t) = -y_1(t) \int \frac{g(t)y_2(t)}{W(t)} dt + y_2(t) \int \frac{g(t)y_1(t)}{W(t)} dt$$

Make sure to get to **standard form** to find the correct $g(t)$!

Notice how $W(t) \neq 0$, guaranteeing a unique solution.

Once you get an answer, be careful to select ONLY the terms that **don't already appear** in the complementary solution!

Variable Coefficients with y_1 Given

ROO - Reduction of Order

Method to solve 2nd order linear with non-constant coefficients.

1. You *must* be given or know the first solution y_1 .
2. Set $y_2 = v(t)y_1(t)$ and differentiate this twice.
3. Substitute the resulting derivatives and the original into the differential equation.

4. **The terms containing $v(t)$ always cancel out (sum to 0).** If not, check your work. Factoring often recreates the homogeneous equation (which equals 0) multiplying $v(t)$.
5. Now the equation only contains first and second derivatives, so you can define $w(t) = v'(t)$, turning the equation into a **first order linear** equation! (Hence, reduction of order).
6. Solve it using the integrating factor or separation of variables.
7. Make sure to return to y as your variable: $w(t) \rightarrow v(t) \rightarrow y_2(t)$.

Springs - Physical Application

IVP given by:

$$my'' + \gamma y' + ky = f(t)$$

$$y(0) = A, y'(0) = B$$

where m is mass, γ is coefficient of friction, k is spring constant, and f is forcing function
All three coefficients are assumed to be **positive**.

Free Vibrations

No damping ($\gamma = 0$)

No external force ($f(t) = 0$)

$$my'' + ky = 0$$

Solution

$$y = A \cos \omega_0 t + B \sin \omega_0 t = R \cos(\omega_0 t - \delta)$$

(Simple harmonic motion)

$R = \sqrt{A^2 + B^2}$	$A = R \cos \delta$
$\delta = \arctan (B/A)$	$B = R \sin \delta$

Be careful: arctan range restricted between $-\pi/2$ to $\pi/2$, so add or subtract π as needed

Damped, Free Vibrations

No external force ($f(t) = 0$)

$$my'' + \gamma y' + ky = 0$$

Three cases, based on the discriminant of r :

$\gamma^2 - 4ac > 0$ (2 distinct, real roots) Relatively high damping Overdamped	
$\gamma^2 - 4ac = 0$ (repeated roots) Medium damping Critically damped	
$\gamma^2 - 4ac < 0$ (2 complex roots) Relatively low damping Underdamped or quasi-periodic	

Damp-free, Forced Vibrations

No damping ($\gamma = 0$)

A periodic forcing function on the right:

$$my'' + ky = F_0 \cos \omega t$$

The solution:

$$y = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t + \frac{F_0 \cos \omega t}{m(\omega_0^2 - \omega^2)}$$

If $\omega \neq \omega_0$, then **beats** occur, which consists of a wave with an oscillating amplitude.

Beats is also known as **amplitude modulation**.

If $\omega = \omega_0$, then **resonance** occurs, which can break the system.

For the resonant case:

$$Y = \frac{F_0}{2\omega_0 m} t \sin \omega_0 t$$

This represents a linearly increasing amplitude

Damped, Forced Vibrations

There will NOT be any resonance since $\gamma > 0$ will eventually keep it in check
At a glance (this case isn't covered in-depth):

$$Y = A \cos \omega t + B \sin \omega t$$

$$A = \frac{F_0(k - \omega^2)}{\Delta^2}, B = \frac{\gamma \omega F_0}{\Delta^2}$$

$$\Delta^2 = (k - m\omega^2)^2 + \gamma^2 \omega^2$$

End of Midterm 2 content

Variable Coefficients Without y_1 Given

$$y'' + p(x)y' + q(x)y = 0$$

We will *only* deal with polynomial coefficients.
We'll only consider $x_0 = 0$ (about 0) to simplify our work.

Power Series

About $x = 0$:

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$$

A Taylor Series is a type of power series.

Taylor Series to Memorize

e^x	$\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ Converges for all x
$\sin x$	$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$

	Converges for all x
$\cos x$	$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$ Converges for all x
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$ Converges only for $x < 1$

Sigma Manipulation

Shifting indices (to bring x to the desired power):

$$\sum_{n=0}^{\infty} a_n x^{n+1} = \sum_{n=1}^{\infty} a_{n-1} x^n$$

Sifting elements out (to get the sigma to the desired starting index):

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + \sum_{n=2}^{\infty} a_n x^n$$

Theorem (Analytic)

If $p(x)$, $q(x)$ are analytic (i.e. have Taylor Series representations / are infinitely differentiable), then the solution to the IVP is analytic.

Ordinary Points

A point at which $p(x)$, $q(x)$ are analytic is an **ordinary point**.

Singular Points

A point at which $p(x)$, $q(x)$ are NOT analytic is a **singular point**.

Regular Singular Points

A singular point at which $x p(x)$, $x^2 q(x)$ are analytic is a **regular singular point**. This means that it has a fairly low amount of “severity” (up to x^2).

Irregular Singular Points

All other types of singular points (outside of the scope of this class).

Recurrence Relation

An equation that helps find all the remaining coefficients of a power series, e.g.

$$a_{n+1} = \frac{a_n}{n+1}, n \geq 1$$

Solving at Ordinary Points

1. Always start with the power series for y :

$$y = \sum_{n=0}^{\infty} a_n x^n$$

2. Differentiate to get y' and y'' , and substitute into the differential equation.
3. Get all the terms to a single power of x by **shifting** indices.
4. Get all sigmas to start at the same n by **sifting** terms out of the sigmas.
5. Set the sum of the terms of each power of x to 0 (this will get a **recurrence relation**).
6. Find the first few terms until either the series can be identified as a known function, or simply left written as a series. The coefficients a_0 and a_1 are arbitrary, serving as the constants of the general solution.

Indicial Equation

An equation that helps find each coefficient of a power series for **regular singular points**, e.g.

$$a_0[2r^2 + r - 1] = 0$$

$$a_n[2(n+r)^2 + (n+r) - 1] = 0$$

Euler's Equation

Simplest situation with regular singular points.

$$x^2 y'' + bxy' + cy = 0$$

Seek solutions in the form of $y = x^r$.

(This is a very similar process to constant coefficient differential equations).

<i>2 Real, Distinct Roots</i>	$y = c_1 x^{r_1} + c_2 x^{r_2}$
<i>Repeated Roots</i>	$y = c_1 x^r + c_2 x^r \ln x $
<i>Complex Roots</i>	$y = x^\lambda [A \cos(\mu \ln x) + B \sin(\mu \ln x)]$ $r = \lambda \pm \mu i$

Note: there *will* be at least one solution that is NOT analytic at $x = 0$.
When solving an IVP in the complex case, do not forget to use the product rule!

Solving at Regular Singular Points

For regular singular points in non-Euler equations, seek solutions in the form of

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

1. We *must* always insist that $a_0 \neq 0$ (it is *always* arbitrary and becomes the constant of one of the fundamental solutions). This helps find the values of r .
2. There will be up to 2 values of r , one for each fundamental solution.
3. Use these values of r in the **indicial equations** of a_1 to a_n .
4. When the roots differ by an integer, be aware of:
 - a. $0 = 0$ (some a_n is arbitrary)
 - b. $0 = (\text{something} \neq 0)$ (some a_n CANNOT be determined)

If $r_2 \neq r_1 + N$, where N is natural,	we will <i>always</i> find 2 solutions. But you must go through the work with both r_1 and r_2 to get both fundamental solutions.
If $r_1 = r_2$,	use Reduction of Order to find the 2nd solution (always possible since the first one is found). The solution is NOT simply $y_2 = y_1 \ln x$. Rather, $y_2 = y_1 \ln x + \sum_{n=0}^{\infty} b_n x^{n+r}$ (some series added at the end). But this situation is out of the scope of this class! (Just FYI)
If $r_2 = r_1 + N$, where N is natural,	it will cause a division by 0 at a_N, since the denominator will become Euler's Characteristic Equation (which equals 0). Two routes:

	1. Substitute the larger root, r_2 (<i>guaranteed</i> exactly one solution; use ROO to find the 2nd). 2. Substitute the smaller root, r_1, (gamble): a. $a_N\{0\} = a_{N-1}\{\neq 0\}$ Dead end (no way to get a_N) b. $a_N\{0\} = a_{N-1}\{0\}$ a_N is arbitrary (along with a_0, as always) (identically satisfied) Gets <i>both</i> solutions simultaneously! <i>Only</i> using the smaller r. But, we will be told which root (the smaller or larger) to substitute to avoid wasting time potentially running into case 2a). For case 2b), make sure to check the $0 = 0$ case before canceling out any factors in the recurrence relation!
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Laplace Transforms

Useful for solving a *class* of IVP (not the solutions themselves).
This class involves **discontinuous forcing functions** and **impulsive forcing**.

The Laplace Transform is an operator (integral) to transform a differential equation to an algebraic one, usually taking it from a domain in t to a domain in s .

It is defined as (script L):

Given $f(t)$, where $t \geq 0$,

$$L\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

Not every $f(t)$ has a Laplace Transform.

Linearity of Laplace Transforms

$$L\{c_1 f + c_2 g\} = c_1 L\{f\} + c_2 L\{g\}$$

Partial fractions and completing the square will be frequently used for this section.

Step Function

Models a switch that turns on at $t = c$

$$u_c(t) = 0 \{t < c\}, 1 \{t \geq c\}$$

Step function are useful when using Laplace Transforms

But their piecewise representation is more useful when sketching

We must know how to go both ways (step function to piecewise, and vice versa)

Impulsive Forcing

A function that is 0 everywhere except a single point.

Physical scenarios: slapping a spring really hard, the force spikes for an instant, but still sets the spring in motion.

Impulse Function

The impulse function is 0 for the entire domain except within τ units of t_0 .

$$d_\tau(t - t_0) = 1/2\tau \{t_0 - \tau < t < t_0 + \tau\}, 0 \{otherwise\}$$

The area under the curve is always 1.

Dirac-Delta Function

The Dirac-Delta Function is just the Impulse Function squeezed to a width of 0:

$$\delta(t - t_0) = \lim_{\tau \rightarrow 0} d_\tau(t - t_0) = 0 \{t \neq t_0\}, \infty \{t = t_0\}$$

The area under the curve is 1:

$$\int_{-\infty}^{\infty} \delta(t - t_0) dt = 1$$

Table of Laplace Transforms

These will be *given* on future exams.

But, we might be asked to find some Laplace Transforms manually to exercise the definition.

$f(t)$	$F(s)$
1	$1/s$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\cos at$	$\frac{s}{s^2+a^2}$
$\sin at$	$\frac{a}{s^2+a^2}$
$f'(t)$	$sF(s) - f(0)$
$f''(t)$	$s^2F(s) - f'(0) - sf(0)$
$u_c(t)$	e^{-cs}/s
$u_c(t)f(t-c)$	$e^{-cs}F(s)$
$e^{ct}f(t)$	$F(s-c)$
$\delta(t-c)$	e^{-cs}

End of Midterm 3 material

Convolution Integral

Remember that

$$L^{-1}\{F(s)G(s)\} \neq f(t)g(t)$$

So, make to make it easier to inverse Laplace transform, we have the convolution integral:

$$L^{-1}\{F(s)G(s)\} = \int_0^t f(t - \tau)g(\tau)d\tau = \int_0^t f(\tau)g(t - \tau)d\tau$$

Which function gets τ and which gets $t - \tau$ does not matter.

System of First Order Linear ODEs

Fact

Any n^{th} -order linear ODE can be written as a system of n 1st order linear ODEs.

For example:

2nd order ODEs can be turned into a system of equations with 2 unknowns by introducing a dummy variable z :

$$my'' + \gamma y' + ky = 0$$

$$\begin{aligned} y' &= z \\ z' &= -(\gamma/m)z - (k/m)y \end{aligned}$$

Similarly, a system of equations with 3 unknowns for 3rd order ODEs with dummy variables z and w :

$$y''' + ay'' + by' + cy = 0$$

$$\begin{aligned} y' &= z \\ z' &= w \\ w' &= -aw - bz - cy \end{aligned}$$

Some Matrix Tidbits

Identity Matrix

The identity matrix, $\vec{I}$, is a square matrix composed of 1s on the main diagonal and 0s everywhere else. It can be any dimension as long as it is square.

Just like multiplying by the scalar 1, the identity matrix does not change another square matrix:

$$\vec{A} \cdot \vec{I} = \vec{A} = \vec{I} \cdot \vec{A}$$

Mapping, Briefly

A square matrix multiplied by a vector (e.g. $\vec{x}$) results in another vector (e.g. $\vec{b}$). (Not the other way around; remember that matrix multiplication is NOT commutative.)

$$\vec{A} \cdot \vec{x} = \vec{b}$$

Eigenvalues & Eigenvectors

Continuing from above, it is particularly interesting when the resulting vector is simply a scaled version of the original vector, without changing direction.

$$\vec{A} \cdot \vec{x} = \lambda \vec{x}$$

The scale factor λ is the **eigenvalue**, and the vector $\vec{x}$ is called the **eigenvector**. ($\vec{A}$ is just a square matrix that is often given to us).

Solving ODEs with Matrices

We will only deal with 2x2 matrices in this class (up to 2nd order) and only the homogeneous kind. The form is as follows:

$$\vec{x}' = \vec{A} \cdot \vec{x} \quad [1]$$

where $\vec{A}$ is given to us, $\vec{x}$ is the solution vector to solve for, and $\vec{x}'$ is its derivative.

We will only deal with constant matrices (those with only constants inside of them) in this class. Plus, since this 2x2 case involves a 2nd order linear differential equation, we can start by assuming the form

$$\vec{x} = \vec{\xi} e^{rt}$$

where $\vec{\xi}$ is a vector to ensure that $\vec{x}$ is a vector too.

Equation [1] always has the trivial solution of $\vec{x} = \vec{0}$. But, we will find the non-trivial solution if and only if

$$[2]$$

$$\det(\vec{A} - \lambda \vec{I}) = 0$$

where λ is an eigenvalue, and $\vec{A}$ is the matrix given in the problem.

This equation will yield a quadratic equation, giving two values of λ . (It will be a cubic equation for a third order ODE, and so on.)

Substitute each λ value into the system. This yields two equations relating x_1 and x_2 , but the relationships are the same. (Use this to check your work; if they don't match up, you might have done something wrong.)

The relationship will help write the two vectors that form the general solution. For example:

$$\begin{aligned} 3x_1 + 3x_2 &= 0 \\ -2x_1 - 2x_2 &= 0 \end{aligned}$$

Both equations give $x_2 = -x_1$ as the relationship, leading to the vector

$$(1; -1)$$

and the general solution is

$$\vec{x} = c_1(1; -1) + \text{second solution}$$

Just like with any quadratic equation, there are three cases for the eigenvalues:

<u>2 Real, Distinct Roots</u>	$\vec{x} = c_1 \vec{\xi}_1 e^{r_1 t} + c_2 \vec{\xi}_2 e^{r_2 t}$
<u>Repeated Roots</u>	We will not cover this case in this class.
<u>2 Complex Roots</u>	
In summary, the general solution is $\vec{x} = c_1 \vec{u} + c_2 \vec{v}$. where the column vectors $\vec{u}$ and $\vec{v}$ are the real and imaginary components, respectively. This complex case requires you to just deal with one of the eigenvectors since the other is simply the complex conjugate (change $i \rightarrow -i$). A more complete explanation:	

Start off $\vec{x}_1$ as a product of the eigenvector $\vec{\xi}_1$, e^{at} , and $\cos bt + i \sin bt$, where the eigenvalues are $a \pm ib$, for example:

$$\vec{x}_1 = [1; -1 - i] e^{-t/2} * (\cos t + i \sin t)$$

(The expression in brackets is a column vector.) (Do not forget the i in front of the sine!)

In this case, the eigenvalues are $-1/2 \pm i$. Simply multiply the sines and cosines into the column vector while separating the real and complex parts to arrive at:

$$\begin{aligned} \vec{x}_1 &= e^{-t/2} [\cos t; -\cos t + \sin t] \\ &+ i e^{-t/2} [\sin t; -\cos t - \sin t] \end{aligned}$$

The first column vector of reals is denoted as $\vec{u}$ and the second column vector (that is multiplied by i) is denoted as $\vec{v}$. The general solution is in the form of:

$$\vec{x} = c_1 \vec{u} + c_2 \vec{v}$$

because $\vec{x}_1 = \vec{u} + i\vec{v}$ and $\vec{x}_2 = \vec{u} - i\vec{v}$. The old constants combine to create c_1 and c_2 . Notice that both fundamental solutions are complex conjugates of each other.

Just like with the complex case for constant coefficients, i can be grouped into the constants, removing it from the general solution.

The duplicate relationship in the system of equations for each root is not so obvious, so you may have to multiply by the complex conjugate to verify. For example:

$$\begin{aligned} (1 - i)\xi_1 - 2\xi_2 &= 0 \\ \xi_1 - (1 + i)\xi_2 &= 0 \end{aligned}$$

Multiply the first equation by the conjugate, $(1 + i)$, to get

$$2\xi_1 - 2(1 + i)\xi_2 = 0 = \xi_1 - (1 + i)\xi_2$$

which is just the second equation.

Luckily, for this complex case, you only have to calculate one root: the other root is simply the complex conjugate.

Make sure to separate the real and imaginary parts to get $\vec{u}$ and $\vec{v}$ vectors. Then, the general solution is just $\vec{x} = c_1 \vec{u} + c_2 \vec{v}$.

Boundary Value Problems

For a differential equation such as

$$y'' + p(x)y' + q(x)y = 0$$

boundary value problems (BVPs) differ from IVPs in that they give values at two different points (usually for y , but sometimes y' instead).

For example, to the differential equation above, the following are initial conditions:

$$y(x_0) = 0$$

$$y'(x_0) = 0$$

Whereas, the following are boundary conditions:

$$y(x_0) = 0$$

$$y(x_1) = 0$$

Notice that x values given are the **same** for **initial** conditions but are **different** for **boundary** conditions. Notice also that the initial conditions usually give y and y' at a point, whereas boundary conditions usually give y at different points (or sometimes y' at different points, or sometimes even a mixture of both).

Solving the differential equation for the general solution is no different. The only new part is imposing the boundary conditions. Unlike IVPs, BVPs can have interesting cases:

Types of Solutions

Case	Example
<u>One Distinct Solution</u>	$y'' + y = 0, y(0) = 1, y(\pi/2) = 0$ Solution: $y = \cos x$

<u>No Solution</u>	$y'' + 4y = 0, y(0) = 1, y(\pi/2) = 0$ No solution!
<u>Infinitely Many Solutions</u>	$y'' + 4y = 0, y(0) = 1, y(\pi/2) = -1$ Solution: $y = \cos 2x + c_2 \sin 2x$ for any arbitrary c_2 !
<u>Homogeneous Problems with Non-Trivial Solutions</u>	$y'' + 4y = 0, y(0) = 0, y(\pi/2) = 0$ Trivial solution: $y = 0$ ("clearly!") Non-trivial solution: $y = c_2 \sin 2x$

Eigenvalues and Eigenfunctions

Take the example:

$$\begin{aligned} y'' + \lambda y &= 0 \\ y(0) &= 0, y(\pi/2) = 0 \end{aligned}$$

The **eigenvalues** are the values of λ for which there are **non-trivial** solutions. These non-trivial solutions are the **eigenfunctions**. Each eigenvalue has a corresponding eigenfunction.

To find all the eigenvalues, consider all the possibilities:

$$\begin{aligned} \lambda &= -\mu^2 < 0 \\ \lambda &= 0 \\ \lambda &= \mu^2 > 0 \end{aligned}$$

(We won't deal with complex eigenvalues for BVPs in this class.)

Simply substitute the respective values for λ in each case. The μ^2 is convenient by avoiding square roots when using the 2nd order constant coefficient characteristic equation.

For this example:

- [Case $\lambda = -\mu^2 > 0$] yields no eigenvalues (you get $\mu = 0$, which is another case)
- [Case $\lambda = 0$] $\lambda = 0$ is NOT an eigenvalue (yields only a trivial solution)
- [Case $\lambda = \mu^2 > 0$] $\lambda = 4n^2$ for $n = 1, 2, \dots$

Answer: $\lambda = 4n^2$ for $n = 1, 2, \dots$

Fourier Series

Fact

Many functions on the **symmetric** interval $[-L, L]$ can be represented by an infinite series of **sines** and **cosines**, i.e., a **Fourier Series**:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

After some work, you can get this useful summary that helps us calculate a_n and b_n :

- $\int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = 0 \{m \neq n\}, L \{m = n\}$
- $\int_{-L}^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx = 0 \{m \neq n\}, L \{m = n\}$
- $\int_{-L}^L \sin \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = 0$

Remember: mixed integrand results in 0 all the time; non-mixed results in L only when both trig arguments are the same, otherwise 0.

This summary leads to the following:

Equation for Coefficients

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx$$

Remember: a involves cosines; b involves sines. The rest of each equation is the same.

Be sure to split the integral appropriately in the case of piecewise functions!

Fun fact: $a_0/2$ is the average value of $f(x)$ (substitute $n = 0$ into the first equation above).

Even & Odd Functions

You can only represent even functions with other even functions (and likewise for odd functions).

If $f(x) = f(-x)$, it is **even**, so it yields a Fourier **cosine** series:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}$$

If $f(x) = -f(-x)$, it is **odd**, so it yields a Fourier **sine** series:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

Product properties of odd and even functions:

- *even func* · *even func* = *even func*
- *even func* · *odd func* = *odd func*
- *odd func* · *odd func* = *even func*

(As a shortcut, you can modify the equations to find a_n and b_n of the cosine and sine Fourier series as so: multiply by $2/L$ instead and set the integral bounds from 0 to L since **in these cases** the integrand is guaranteed to be even, as per the properties above. This may not work when $f(x)$ is neither even nor odd. This was not discussed in class, so only use this shortcut at your own discretion.)

Making an Interval Symmetric

Sometimes we may be given an interval such as $[0, L]$. To get it to the **symmetric** interval $[-L, L]$, we can extend the function conveniently. By making it either odd or even, we only have to deal with a Fourier sine or cosine series instead of a full series.

An extension to create an **even** function:

$$F(x) = f(x) \{0 < x < L\}, f(-x) \{-L < x < 0\}$$

An extension to create an **odd** function:

$$G(x) = f(x) \{0 < x < L\}, -f(-x) \{-L < x < 0\}$$

Both extensions converge to $f(x)$ on $[0, L]$ but may behave differently on $[-L, 0]$.

Updated up to and including Lecture 36 (5/6/2025 (T))

End of the MATH 222 H02 Curriculum for Spring 2025